A Conditional Bidefect Reduction for Aubrun's Partial-Transpose Moment Count

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June 24, 2026

Abstract

We isolate a purely combinatorial reduction in Aubrun's moment method for partially transposed Wishart matrices. The reduction replaces a one-parameter defect count by a two-parameter bidefect count attached to the two Cayley geodesic inequalities for the long cycle. This separation is exactly what is needed for the two analytic bases in the partial-transpose expansion: the plus-defect is paid by a shifted d-base and the minus-defect is paid by a second shifted base, instantiated by the $\sqrt{s}$-scale in the Wishart application. We prove the scalar absorption and the genus-induction reduction in full detail. The final enumerative input is reduced to two familiar ingredients: the genus-zero one-sided geodesic count and a Chapuy-type slicing recurrence for the associated unicellular maps. The main result is a self-contained conditional reduction theorem.

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1 Introduction

Let W be a complex Wishart matrix on $\mathbb{C}^d \otimes \mathbb{C}^d$, and let W^Γ denote the partial transpose on one tensor factor. Aubrun's proof of the sharp PPT threshold studies moments of W^Γ by expanding them into permutation-indexed Wick terms [1]. In the balanced regime $s \asymp d^2$, the leading terms are controlled by Cayley-geodesic permutations. The delicate part is not the rank budget itself; it is the count of the almost-geodesic terms and the way their defects are absorbed by the two different analytic bases.

The purpose of this note is to state and prove the clean reduction that emerges from that bookkeeping. All Cayley lengths below are taken in the Cayley graph of $\mathfrak{S}_n$ generated by transpositions. Write $n = m + 1$, let $\gamma_n = (0\,1\,\cdots\,n-1) \in \mathfrak{S}_n$, and let $\#(\sigma)$ be the number of cycles of a permutation, fixed points included. For $\pi \in \mathfrak{S}_n$, define the two oriented Cayley defects

$$\delta_+(\pi) = n + 1 - (\#(\pi) + \#(\gamma_n\pi)), \qquad \delta_-(\pi) = n + 1 - (\#(\pi) + \#(\pi\gamma_n^{-1})).$$

The central counting object is therefore not a total-defect slice, but the bidefect slice

$$\mathcal{C}_m(a, b) = \#\{\pi \in \mathfrak{S}_{m+1} : \delta_+(\pi) = a,\ \delta_-(\pi) = b\}.$$

The analytic argument requires a bidefect bound of the following shape, made precise in Definition 3.1 and Assumption 3.2:

$$\mathcal{C}_m(a, b) \le 4^{m+1}\binom{m+3}{a}\binom{m+1}{b}\mathcal{Q}_m^{a+b}, \qquad \mathcal{Q}_m = (4(m + 1))^{36},$$

on the analytic ranges $0 \le a \le m+3$, $0 \le b \le m+1$. The binomial factors are not decorative. They are exactly what lets the double sum split into two binomial expansions:

$$\sum_{a,b} \mathcal{C}_m(a, b)d^{m+3-a}S^{m+1-b} \lesssim 4^{m+1}(d + \mathcal{Q}_m)^{m+3}(S + \mathcal{Q}_m)^{m+1}.$$

This is the scalar mechanism needed by the partial-transpose moment estimate. The rest of the note reduces this bidefect bound to two enumerative inputs: the genus-zero count and a Chapuy-type slicing recurrence. The precise form of the reduction is Theorem 4.7.

2 Permutation Defects

Throughout the paper $n = m + 1 \ge 1$, and $\gamma_n = (0\,1\,\ldots\,n-1) \in \mathfrak{S}_n$. We write $\#(\sigma)$ for the number of cycles of σ, including fixed points, and

$$|\sigma| = n - \#(\sigma)$$

for the Cayley length with respect to transpositions.

Definition 2.1 (Oriented defects)**.** *For $\pi \in \mathfrak{S}_n$, define*

$$\delta_+(\pi) = n + 1 - \#(\pi) - \#(\gamma_n\pi), \qquad \delta_-(\pi) = n + 1 - \#(\pi) - \#(\pi\gamma_n^{-1}).$$

The total defect is

$$\Delta(\pi) = \delta_+(\pi) + \delta_-(\pi).$$

Lemma 2.2 (Non-negativity)**.** *For every $\pi \in \mathfrak{S}_n$,*

$$\delta_+(\pi) \ge 0, \qquad \delta_-(\pi) \ge 0.$$

Proof. The Cayley length is the minimal number of transpositions needed to write a permutation, so it is subadditive by concatenating factorizations. Applying this triangle inequality gives

$$|\pi| + |\pi^{-1}\gamma_n^{-1}| \ge |\gamma_n^{-1}| = n - 1.$$

Since $|\sigma| = n - \#(\sigma)$ and $\#(\pi^{-1}\gamma_n^{-1}) = \#(\gamma_n\pi)$, this is equivalent to

$$\#(\pi) + \#(\gamma_n\pi) \le n + 1.$$

This proves $\delta_+(\pi) \geq 0$. For the minus defect, the same triangle inequality gives

$$|\pi| + |\pi^{-1}\gamma_n| \geq |\gamma_n| = n - 1.$$

Equivalently,

$$(n - \#(\pi)) + (n - \#(\pi^{-1}\gamma_n)) \geq n - 1.$$

Finally

$$\#(\pi^{-1}\gamma_n) = \#((\pi^{-1}\gamma_n)^{-1}) = \#(\gamma_n^{-1}\pi) = \#(\pi\gamma_n^{-1}),$$

where the last equality uses that uv and vu have the same cycle count. Hence

$$\#(\pi) + \#(\pi\gamma_n^{-1}) \leq n + 1,$$

which is exactly $\delta_-(\pi) \geq 0$. $\square$

Definition 2.3 (Aubrun rank)**.** *The permutation rank entering the balanced Wick term is*

$$\mathcal{R}(\pi) = 2\#(\pi) + \#(\gamma_n\pi) + \#(\pi\gamma_n^{-1}).$$

Proposition 2.4 (Rank and bidefect identity)**.** *For every $\pi \in \mathfrak{S}_n$,*

$$\mathcal{R}(\pi) + \delta_+(\pi) + \delta_-(\pi) = 2n + 2.$$

In particular $\mathcal{R}(\pi) \leq 2n + 2$.

Proof. This is immediate from the definitions:

$$\begin{aligned}\delta_+(\pi) + \delta_-(\pi) &= 2n + 2 - 2\#(\pi) - \#(\gamma_n\pi) - \#(\pi\gamma_n^{-1}) \\ &= 2n + 2 - \mathcal{R}(\pi).\end{aligned}$$

The last assertion follows from the non-negativity of the two oriented defects. $\square$

Example 2.5 (The case $n = 3$)**.** *Let $\gamma_n = (0\,1\,2)$. The following sample rows illustrate the identity in Proposition 2.4.*

π	$\#(\pi)$	$\#(\gamma_n\pi)$	$\#(\pi\gamma_n^{-1})$	$\delta_+(\pi)$	$\delta_-(\pi)$	$\mathcal{R}(\pi)$
id	3	1	1	0	0	8
$(0\,1)$	2	2	2	0	0	8
γ_n	1	1	3	2	0	6
γ_n^{-1}	1	3	1	0	2	6

In every row $\mathcal{R} + \delta_+ + \delta_- = 8 = 2n + 2$.

3 The Bidefect Count and Scalar Absorption

The following definition is the exact count required by the shifted-base moment estimate.

Definition 3.1 (Bidefect count)**.** *For $a, b \geq 0$, set*

$$\mathcal{C}_m(a, b) = \#\{\pi \in \mathfrak{S}_{m+1} : \delta_+(\pi) = a,\ \delta_-(\pi) = b\}.$$

Let

$$\mathcal{Q}_m = (4(m + 1))^{36}.$$

The exponent 36 is much larger than the genus induction alone requires; it is retained here because it is the uniform polynomial loss used in the surrounding compatible-couple *estimates. This note does not derive that ambient loss: the scalar argument below only needs a loss parameter large enough to dominate the genus-slicing factor, and it allows any larger replacement $Q \geq \mathcal{Q}_m$. Define the bidefect envelope*

$$\mathcal{E}_m(a,b) = 4^{m+1}\binom{m+3}{a}\binom{m+1}{b}\mathcal{Q}_m^{a+b}.$$

The binomial factors are inserted so that the scalar sum factors into the two shifted bases $(d+Q)^{m+3}$ and $(S+Q)^{m+1}$. Since the reduction below uses only a one-sided plus-defect genus count, the factor $\binom{m+1}{b}\mathcal{Q}_m^b$ is deliberate padding rather than independent minus-defect information.

The cutoffs in the next assumption are dictated by Proposition 2.4. If $\delta_+(\pi) = a$ and $\delta_-(\pi) = b$, then

$$\mathcal{R}(\pi) = 2m + 4 - a - b.$$

In the partial-transpose Wick expansion, the offsets $m+3$ and $m+1$ are the base exponents attached to the two separated parameters before defects are charged. Thus the separated bases have schematic powers

$$d^{m+3-a}S^{m+1-b}.$$

The shifted-base majorant only needs the terms whose displayed exponents are nonnegative. Thus the analytic ranges are precisely

$$0 \leq a \leq m+3, \qquad 0 \leq b \leq m+1.$$

The theorem below is a reduction for this analytic part of the sum; it makes no claim about slices outside these ranges.

Assumption 3.2 (Bidefect count bound)**.** *For every $m \geq 0$, every $0 \leq a \leq m+3$, and every $0 \leq b \leq m+1$,*

$$\mathcal{C}_m(a,b) \leq \mathcal{E}_m(a,b).$$

In the scalar theorem, S is only a placeholder for the second analytic base. In the balanced Wishart application one substitutes the corresponding $\sqrt{s}$-scale, but the absorption argument itself is just the displayed two-variable inequality.

Theorem 3.3 (Shifted-base scalar absorption)**.** *Assume the bidefect count bound for a fixed m. Let $d, S, Q \geq 0$ and $\mathcal{Q}_m \leq Q$. Then*

$$\sum_{a=0}^{m+3}\sum_{b=0}^{m+1}\mathcal{C}_m(a,b)d^{m+3-a}S^{m+1-b} \leq 4^{m+1}(d+Q)^{m+3}(S+Q)^{m+1}.$$

Proof. By Assumption 3.2, the left-hand side is at most

$$4^{m+1}\sum_{a=0}^{m+3}\sum_{b=0}^{m+1}\binom{m+3}{a}\binom{m+1}{b}\mathcal{Q}_m^{a+b}d^{m+3-a}S^{m+1-b}.$$

The double sum factors:

$$\begin{aligned} &4^{m+1}\left(\sum_{a=0}^{m+3}\binom{m+3}{a}\mathcal{Q}_m^a d^{m+3-a}\right)\left(\sum_{b=0}^{m+1}\binom{m+1}{b}\mathcal{Q}_m^b S^{m+1-b}\right) \\ &\quad = 4^{m+1}(d+\mathcal{Q}_m)^{m+3}(S+\mathcal{Q}_m)^{m+1}. \end{aligned}$$

Since $\mathcal{Q}_m \leq Q$ and $d, S, Q \geq 0$, monotonicity of powers gives the claimed bound. □

Remark 3.4. *The proof explains why a one-parameter total-defect envelope is the wrong object for the balanced partial-transpose expansion. The two defects pay two different bases. Collapsing them too early destroys the binomial structure that produces $(d+Q)^{m+3}(S+Q)^{m+1}$.*

4 One-Sided Genus Reduction

The remaining task is to prove Assumption 3.2. The reduction used here passes through the one-sided defect δ_+.

Definition 4.1 (Plus-defect and genus counts)**.** *Let*

$$P_m(a) = \#\{\pi \in \mathfrak{S}_{m+1} : \delta_+(\pi) = a\}.$$

For $g \geq 0$, define

$$G_m(g) = \#\{\pi \in \mathfrak{S}_{m+1} : \delta_+(\pi) = 2g\}.$$

Lemma 4.2 (Parity)**.** *For every $\pi \in \mathfrak{S}_{m+1}$, the plus-defect $\delta_+(\pi)$ is even.*

Proof. Let $n = m + 1$. The sign identity $\operatorname{sgn}(\gamma_n \pi) = \operatorname{sgn}(\gamma_n) \operatorname{sgn}(\pi)$, together with $\operatorname{sgn}(\sigma) = (-1)^{n-\#(\sigma)}$, gives

$$n - \#(\gamma_n \pi) \equiv (n-1) + (n - \#(\pi)) \pmod 2.$$

Rearranging,

$$n + 1 - \#(\pi) - \#(\gamma_n \pi) \equiv 0 \pmod 2.$$

This is exactly $\delta_+(\pi) \equiv 0 \pmod 2$. □

Assumption 4.3 (Genus-zero and slicing inputs)**.** *For every $m \geq 0$:*

(i) the genus-zero count satisfies

$$G_m(0) \leq 4^{m+1};$$

(ii) the genus slices satisfy the Chapuy-type recurrence

$$G_m(g+1) \leq (2(m+1))^3 G_m(g) \qquad (g \geq 0).$$

Proposition 4.4 (Iterated genus bound)**.** *Under Assumption 4.3,*

$$G_m(g) \leq 4^{m+1}(2(m+1))^{3g} \qquad (m, g \geq 0).$$

Proof. Induct on g. The case $g = 0$ is Assumption 4.3(i). If the bound holds at g, then Assumption 4.3(ii) gives

$$G_m(g+1) \leq (2(m+1))^3 G_m(g) \leq 4^{m+1}(2(m+1))^{3(g+1)}.$$

□

Proposition 4.5 (One-sided plus-count envelope)**.** *Under Assumption 4.3, for every $m, a \geq 0$,*

$$P_m(a) \leq 4^{m+1} \mathcal{Q}_m^a.$$

Proof. If a is odd, Lemma 4.2 gives $P_m(a) = 0$. If $a = 2g$, then $P_m(a) = G_m(g)$, and Proposition 4.4 gives

$$P_m(a) \leq 4^{m+1}(2(m+1))^{3g}.$$

The chosen loss satisfies

$$\mathcal{Q}_m^2 = (4(m+1))^{72} \geq (2(m+1))^3.$$

Therefore

$$(2(m+1))^{3g} \leq \mathcal{Q}_m^{2g} = \mathcal{Q}_m^a.$$

The displayed inequality uses far less than the exponent 36; the larger power is kept only to match the ambient Aubrun loss budget. □

Theorem 4.6 (Genus route to the bidefect count)**.** *Assumption 4.3 implies Assumption 3.2.*

Proof. Only the analytic ranges of Assumption 3.2 are being claimed. For fixed a, b in those ranges, the bidefect slice embeds in the plus-defect slice:

$$\{\pi : \delta_+(\pi) = a,\ \delta_-(\pi) = b\} \subseteq \{\pi : \delta_+(\pi) = a\}.$$

Therefore

$$\mathcal{C}_m(a, b) \le P_m(a) \le 4^{m+1} \mathcal{Q}_m^a.$$

On the analytic ranges $0 \le a \le m + 3$, $0 \le b \le m + 1$, the binomial coefficients are at least one, and $\mathcal{Q}_m^a \le \mathcal{Q}_m^{a+b}$. Thus

$$\mathcal{C}_m(a, b) \le 4^{m+1} \binom{m+3}{a} \binom{m+1}{b} \mathcal{Q}_m^{a+b} = \mathcal{E}_m(a, b).$$

$\square$

Theorem 4.7 (Main bidefect reduction)**.** *Under Assumption 4.3, for every $m \ge 0$, every $d, S, Q \ge 0$ with $\mathcal{Q}_m \le Q$, the bidefect sum truncated to the analytic ranges $0 \le a \le m + 3$, $0 \le b \le m + 1$ satisfies*

$$\sum_{a=0}^{m+3} \sum_{b=0}^{m+1} \mathcal{C}_m(a, b) d^{m+3-a} S^{m+1-b} \le 4^{m+1} (d + Q)^{m+3} (S + Q)^{m+1}.$$

Proof. Combine Theorems 4.6 and 3.3. $\square$

Remark 4.8. *The theorem is only a statement over the analytic ranges displayed in the double sum. It makes no assertion about defect pairs with $a > m + 3$ or $b > m + 1$.*

5 Route to the Genus-Zero and Chapuy-Slicing Inputs

This section records the enumerative inputs still needed to discharge Assumption 4.3. It does not prove those inputs. Its purpose is to identify exactly where the remaining combinatorial work lies.

5.1 Genus zero

The condition $\delta_+(\pi) = 0$ is the equality case of the Cayley inequality

$$\#(\pi) + \#(\gamma_n \pi) \le n + 1.$$

Equivalently, π lies on a geodesic between the identity and the long cycle, up to the left/right convention. The required genus-zero input is to identify this equality case with noncrossing partitions of the linearly ordered set $\{0, \ldots, n - 1\}$. The desired bound

$$G_m(0) \le 4^{m+1}$$

is then a Catalan-type estimate: noncrossing partitions of an n-point set are counted by the Catalan number C_n, and $C_n \le 4^n$.

One possible route is to prove noncrossing of the cycle partition and an injectivity statement saying that the ordered predecessor data reconstructs the original permutation. Once that encoding is available, the separate cardinality estimate for noncrossing partitions is elementary.

5.2 Positive genus

For $\delta_+(\pi) = 2g$, the permutation pair $(\pi, \gamma_n\pi)$ should be identified with the standard encoding of a unicellular map of genus g. The missing enumerative input is a slicing map of Chapuy type: from genus $g+1$ one selects boundedly many marked local data and cuts to genus g. The desired coarse form is only

$$G_m(g+1) \le (2n)^3 G_m(g), \qquad n = m+1.$$

This is far weaker than the sharp identities available for unicellular maps [4], but it must be proved in the exact permutation model used by the moment expansion.

6 Relation to the Partial-Transpose Moment Expansion

Aubrun's partially transposed Wishart moment formula assigns to each $\pi \in \mathfrak{S}_{m+1}$ a contribution whose powers of the two large bases are governed by the cycle counts appearing above. The rank identity Proposition 2.4 supplies the total exponent budget

$$\mathcal{R}(\pi) = 2m + 4 - \delta_+(\pi) - \delta_-(\pi).$$

If the two bases are kept separate, a term of bidefect (a, b) has the schematic size

$$d^{m+3-a} S^{m+1-b}$$

up to the universal polynomial loss $\mathcal{Q}_m^{a+b}$ already present in Aubrun's compatible-couple bound. The two exponents are chosen so that their sum recovers exactly the rank identity:

$$(m+3-a) + (m+1-b) = 2m + 4 - a - b = \mathcal{R}(\pi)$$

when $\delta_+(\pi) = a$ and $\delta_-(\pi) = b$. Thus the separated bookkeeping keeps the two analytic bases visible while preserving the total cycle-rank budget. Theorem 4.7 then gives the exact scalar estimate needed by the shifted-base route.

This note does not reprove the whole random-matrix theorem. Its contribution is the clean combinatorial reduction: once the genus-zero count and Chapuy slicing recurrence are supplied in the exact permutation model, the bidefect count and scalar absorption follow with no further probabilistic input.

Verification Checklist for the Reduction

A referee can verify the note by checking the following items.

1. The oriented defects are nonnegative consequences of the two Cayley triangle inequalities.
2. The rank identity is a direct algebraic identity.
3. The scalar bound is exactly the binomial theorem applied twice.
4. The plus-defect is even by the sign identity.
5. The genus induction proves the displayed one-sided count envelope.
6. The bidefect count follows from the one-sided count by inclusion and by the positivity of the two binomial coefficients on the analytic ranges.
7. The only genuine remaining inputs are the genus-zero count and the Chapuy slicing recurrence in the exact permutation model.

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